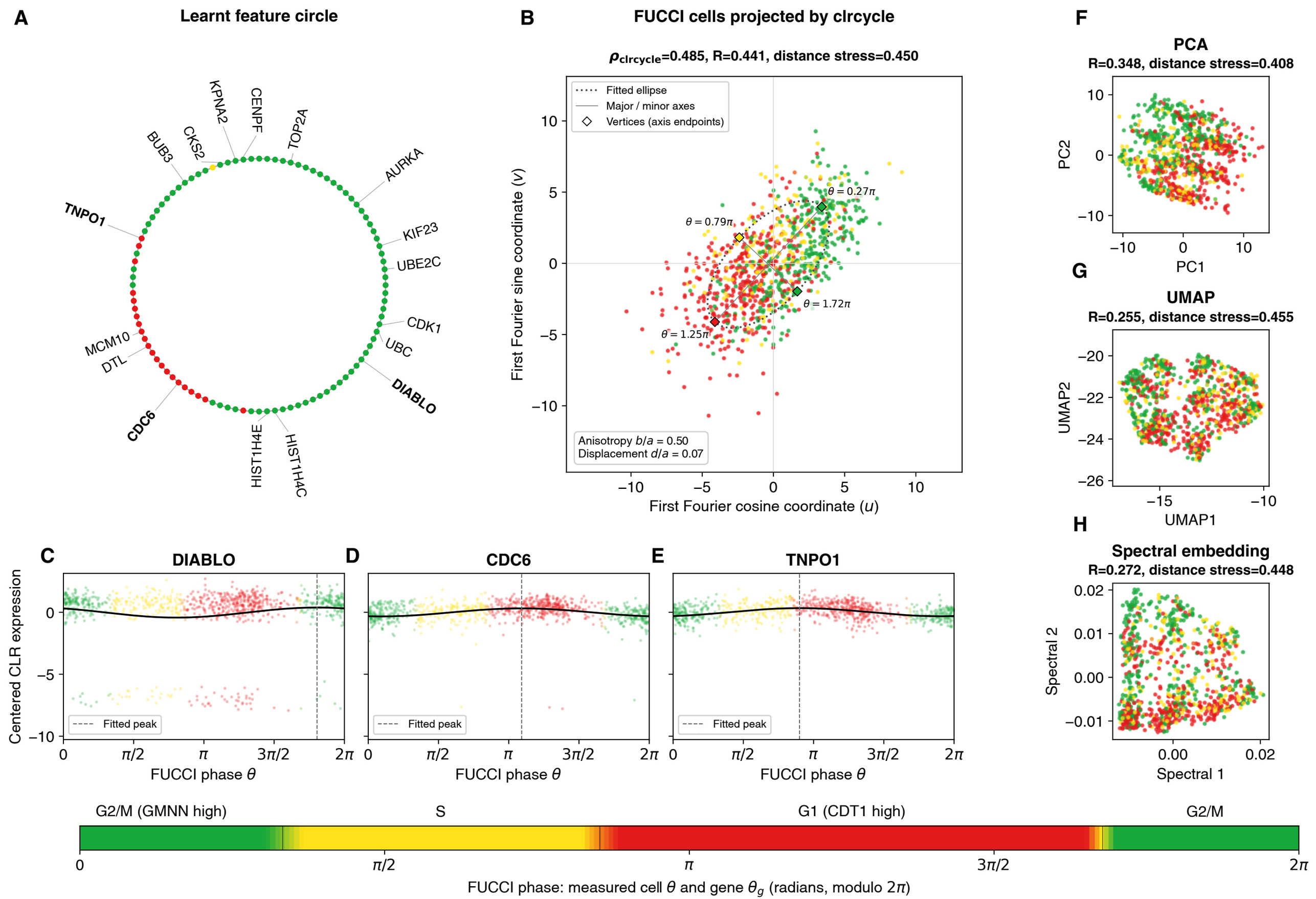

A
Learnt feature circle
KPNA2
CENPF
TOP2A
BUB3
CKS2
AURKA
TNPO1
KIF23
UBE2C
CDK1
MCM10
UBC
DTL
DIABLO
CDC6
HIST1H4E
HIST1H4C
B
FUCCI cells projected by clrcycle
ρclrcycle=0.485, R=0.441, distance stress=0.450
Fitted ellipse
Major / minor axes
Vertices (axis endpoints)
θ = 0.27π
θ = 0.79π
θ = 1.72π
θ = 1.25π
Anisotropy b/a = 0.50
Displacement d/a = 0.07
First Fourier sine coordinate (v)
First Fourier cosine coordinate (u)
F
PCA
R=0.348, distance stress=0.408
PC2
PC1
G
UMAP
R=0.255, distance stress=0.455
UMAP2
UMAP1
H
Spectral embedding
R=0.272, distance stress=0.448
Spectral 2
Spectral 1
C
DIABLO
D
CDC6
E
TNPO1
Centered CLR expression
Fitted peak
FUCCI phase θ
G2/M (GMNN high)
S
G1 (CDT1 high)
G2/M
FUCCI phase: measured cell θ and gene θg (radians, modulo 2π)